Is Graph Bipartite? (medium)

Problem Statement

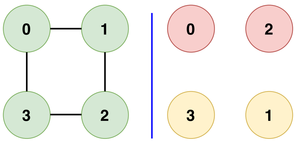
Given a 2D array `graph[][]`, representing the undirected graph, where `graph[u]` is an array of nodes that are connected with node `u`.

Determine whether a given undirected graph is a bipartite graph.

The graph is a bipartite graph, if we can split the set of nodes into two distinct subsets such that no two nodes within the same subset are adjacent (i.e., no edge exists between any two nodes within a single subset).

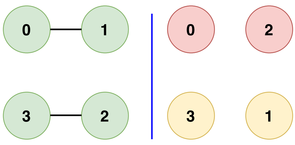
Examples

1. **Example 1**:
   - **Input**: `[[1,3], [0,2], [1,3], [0,2]]`



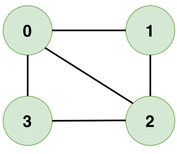
- **Expected Output**: `true`
- **Justification**: The nodes can be divided into two groups: {0, 2} and {1, 3}. No two nodes within each group are adjacent, thus the graph is bipartite.

2. **Example 2**:
   - **Input**: `[[1], [0], [3], [2]]`



- **Expected Output**: `true`
- **Justification**: The graph is a simple chain of 4 nodes. It's clearly bipartite as we can have {0, 2} and {1, 3} as the two subsets.

3. **Example 3**:
   - **Input**: `[[1,2,3],[0,2],[0,1,3],[0,2]]`



- **Expected Output**: `false`
- **Justification**: We found that edges (1, 2), (1, 3), and (2, 3) connect nodes within the same set, which violates the condition for a bipartite graph where each edge must connect nodes in different subsets. Thus, there's no way to divide this graph into two subsets that satisfy the bipartite condition.

Try it yourself

Try solving this question here:

Java

```java
public class Solution {
    public boolean isBipartite(int[][] graph) {
        // ToDo: Write Your Code Here.
        return false;
    }
}
```

Test Case | Run Code Result | Stdout

[[1],[0,2],[1]]

Use Example Testcases | Run | Submit